How Accurate Is Regression Adjustment for the Global Treatment Effect Under Network Interference?

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Abstract

Regression adjustment, which regresses outcomes on a unit's own treatment and a summary of its neighbours' treatments and then predicts under all-treated and all-control, is a common way to estimate the global average treatment effect (GATE) under network interference, but little is known about how accurate it is or what limits it. We study it on synthetic outcomes with an exactly known GATE on four synthetic graph families and two public Facebook graphs, over 72 main-grid cells (graphs $\times$ interference mechanisms $\times$ designs) and eleven sweep experiments, separating bias from variance and reporting bootstrap uncertainty. With a correct linear-exposure model the estimator is unbiased and its error falls as $n^{-0.51}$, but in our default SBM setting (mean degree 20) it beats the difference in means only for $n \gtrsim 2{,}000$ and interference at least as strong as the direct effect, and this break-even moves with degree and confounding; its error is variance-dominated and grows with mean degree and community-level confounding (relative RMSE 0.09 to 2.51 for mean degree 5 to 160). With a saturating or threshold response or two-hop interference it is biased by amounts that do not shrink with n (e.g. a relative bias of +0.53 under a saturating response at $p = 0.1$), and errors in the graph bias it towards zero; a first-order formula predicts the size of the bias in the linear-exposure family; flexible learners trade bias for variance without dominating; and graph-cluster randomisation removes most of the advantage over the difference in means. Nominal 95% cluster-robust intervals cover 0.85–0.94 even when the estimator is unbiased. A data-computable diagnostic of exposure support that we expected to predict the bias works only for one response family. All findings concern simulated outcomes.

1 Introduction

Randomised experiments on networks, such as platform experiments with social, marketplace or communication graphs, violate the no-interference assumption: a unit's outcome depends on the treatment of its neighbours. The usual target is the *global average treatment effect* (GATE), the average outcome difference between treating everybody and treating nobody. The classical design-based answers (exposure mappings with Horvitz–Thompson or Hájek weights, graph-cluster randomisation) are unbiased under a correctly specified exposure model, but they need units to be fully exposed with non-negligible probability, which is rare in dense graphs, and they can have very large variance [Aronow and Samii, 2017, Ugander et al., 2013, Eckles et al., 2017]. A popular practical alternative is *regression adjustment*: regress outcomes on one's own treatment and a summary of the neighbours' treatment, typically the treated fraction of neighbours, then predict the outcome with everybody treated and with nobody treated [Chin, 2019, Han and Ugander, 2023]. It uses every unit, is cheap and has simple standard errors.

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What is not well understood is *how accurate it is, what drives its error, and when it breaks*. The prediction points lie far from where the design puts the data, so the estimator is an extrapolation whose accuracy depends on the design, the graph and the response shape; the cost of flexibility is variance. A practitioner who wants to trust or discard such an estimate needs to know which of these dominates in her setting.

We answer these questions in a controlled study with a known estimand. Outcomes are synthetic, so the GATE is exact, and are generated on four synthetic graph families and two public Facebook graphs; we vary the interference mechanism, the experimental design, graph size and density, interference strength, noise, the quality of the graph and the model class. Because every estimator sees the same assignments, differences are paired. Our contributions are:

- A reproducible benchmark of 72 main cells (6 graphs $\times$ 4 mechanisms $\times$ 3 designs) plus sweeps over eleven factors, with bootstrap uncertainty for every plotted number and a second seed set for replication (Section 4, Appendix).
- An error decomposition into approximation (bias from the model class and the design's exposure support, measured with noise-free population fits) and estimation error, with three first-order analytic predictions (P1–P3) for bias and variance in the linear-exposure family that we check numerically on several hundred graph draws.
- Findings with their scope: under a correct specification the estimator is unbiased and clearly better than the difference in means only for sufficiently large n and sufficiently strong interference; its error is variance-dominated and grows with degree and with community-level confounding; with nonlinear exposure, wrong feature maps or graph errors it is biased by an amount that does not vanish with n, and flexible learners trade this bias for variance without dominating; cluster randomisation removes most of the advantage over the difference in means; and the standard cluster-robust intervals under-cover even when the point estimate is unbiased.
- Negative results: a support diagnostic we expected to predict the bias works for one response family only; multi-p designs and coverage trends under misspecification were not run.

All conclusions are about simulated outcomes on these graphs; we do not claim that they transfer to a given platform (Section 5).

2 Related Work

Estimating the global effect under interference. Randomisation-based inference under interference was developed around *exposure mappings*, which reduce the full assignment vector to a coarser exposure; Horvitz–Thompson and Hájek estimators of exposure-specific means then follow from known exposure probabilities [Aronow and Samii, 2017]. For social networks, graph cluster randomisation increases the probability that a unit's whole neighbourhood is treated, and the resulting HT estimators have a bias–variance trade-off governed by the clustering [Ugander et al., 2013, Eckles et al., 2017]; Karrer et al. [2021] describe such a system in production. Leung [2022] studies cluster designs and HT estimation of the global effect when interference decays with distance, and Cortez et al. [2022] show that staggered rollouts permit unbiased estimation of the total effect under low-order interactions without knowledge of the network. Chen et al. [2023] derive bias and variance of the HT estimator under one-hop interference and optimise the covariance of the assignment. Under unknown interference, standard estimators can still target a generalised estimand that averages over spillovers [Sävje et al., 2021], and Yu et al. [2022] address estimation of the total effect when the network is unknown.

Regression-based estimators. The estimator studied here fits an outcome model on the assignment and graph-derived features from a single experiment and then predicts outcomes under all-treated and all-control assignments. Chin [2019] proposed this regression adjustment for Bernoulli designs, gave properties in linear models together with resampling-based inference, and evaluated it on a Facebook100 network. Han and Ugander [2023] generate and select graph- and treatment-based covariates, and construct block-bootstrap confidence intervals. These works show that regression adjustment can substantially reduce the bias of the difference in means when the model is adequate; they place more emphasis on the correctly specified and mildly misspecified linear regime than on characterising what determines the estimation error and where it breaks down, which is our focus.

3 Problem Setup and Estimators

3.1 Estimand and regression estimators

Units $i = 1, \dots, n$ sit on a fixed undirected graph with adjacency A, degrees d_i and row-normalised adjacency $W = D^{-1}A$. A design draws an assignment $\mathbf{z} \in \{0,1\}^n$; the outcome is $Y_i = \mu_i(\mathbf{z}) + \varepsilon_i$ with a mean function μ_i that may depend on all of $\mathbf{z}$ and mean-zero noise ε_i. The estimand is the global average treatment effect

$$\tau = \frac{1}{n}\sum_{i=1}^{n}\big[\mu_i(\mathbf{1}) - \mu_i(\mathbf{0})\big].$$

Since outcomes are synthetic, τ is computed *exactly* from the noise-free mean function; there is no Monte-Carlo error in the target.

A *regression-based* estimator fits a model $\hat{f}$ from features $x_i(\mathbf{z})$ built from the assignment and the graph to the observed outcomes of one experiment, and returns

$$\hat{\tau} = \frac{1}{n}\sum_{i=1}^{n}\big[\hat{f}\big(x_i(\mathbf{1})\big) - \hat{f}\big(x_i(\mathbf{0})\big)\big].$$

The key structural fact is that the two prediction points $x_i(\mathbf{1}), x_i(\mathbf{0})$ generally lie outside the support of the features observed under the design: under Bernoulli(p) randomisation the treated-neighbour fraction $f_i = (W\mathbf{z})_i$ concentrates around p, while the predictions need $f_i = 1$ and $f_i = 0$. We study the following feature maps (all include the intercept and z_i): `ols_f` adds f_i (the setting of Chin 2019); `ols_cnt` replaces it by the scaled count $(A\mathbf{z})_i/\bar{d}$; `ols_f2` adds the two-hop fraction $(W^2\mathbf{z})_i$; `ols_poly` adds f_i^2 and $z_i f_i$; `ols_f_deg` adds log-degree and its interactions; `ridge_rich` fits a ridge regression on a rich set of such features; and `gbm` is a histogram gradient-boosted tree regressor on $(z_i, f_i, (W^2\mathbf{z})_i, \text{log-degree}, \text{clustering})$. Baselines are the difference in means (DM) and a Hájek estimator contrasting units whose whole closed neighbourhood is treated versus control, with inverse probabilities of that event under the design [Aronow and Samii, 2017, Ugander et al., 2013].

3.2 Error decomposition

For each estimator we separate (i) the *approximation error*, the error of the estimator when it is fit to the noise-free, baseline-free mean function $\mu(\mathbf{z})$ on the same assignment (a "population" fit), which isolates what the model class can do from the design's exposure support, from (ii) the *estimation error* due to finite n, baselines and noise. Across replicates we report bias, standard deviation and RMSE, with RMSE relative to $|\tau|$ as the scale-free primary metric: the question is how much of the error is systematic (does not shrink with n) and how much is variance.

3.3 Analytic predictions in the linear-exposure family

We derive first-order predictions for the OLS(z, f) estimator under Bernoulli(p) and compare them to simulation (Section 4). We stress that these are large-n linearisation arguments checked numerically, not finite-sample theorems. Write the mean function as $\mu_i(\mathbf{z}) = b_i + \delta z_i + g(f_i)$ with baselines b_i and direct effect δ, so that $\tau = \delta + g(1) - g(0)$.

P1 (nonlinear exposure response). Because z_i is independent of f_i, the population OLS slope on f is $\beta_f^* = \mathrm{Cov}(g(f), f)/\mathrm{Var}(f)$ with $f_i \mid d_i \sim \mathrm{Bin}(d_i, p)/d_i$, and the asymptotic bias of $\hat{\tau}$ is $\beta_f^* - [g(1) - g(0)]$. It vanishes for linear g; for smooth g and high degree $\beta_f^* \to g'(p)$, so the bias tends to a tangent-versus-secant gap that does not shrink with n.

P2 (variance, correct specification). With a fixed centred baseline vector b and noise variance σ_e^2, $\mathrm{Var}(\hat{\tau}) \approx \big[\|b + d_H W^\top b\|^2/n + \sigma_e^2(1 + d_H)\big]/\big(np(1-p)\big)$, where d_H is the harmonic mean degree. The error is design-based (the baselines projected on random exposures contribute), grows with degree, and is independent of the interference strength.

P3 (linear in z but not in f). If $\mu_i(\mathbf{z}) = \delta z_i + \sum_j M_{ij} z_j$ (heterogeneous spillovers, two-hop terms, count-scaled exposure), the OLS limit has z-slope $\delta + \mathrm{mean}_i M_{ii}$ and f-slope $\sum_{ij} W_{ij} M_{ij} / \sum_{ij} W_{ij}^2$. For $M = \mathrm{diag}(\gamma) W$ the f-slope is a $1/d_i$-weighted mean of γ_i

whereas τ weights units equally, so the estimator is biased whenever spillover strength correlates with degree.

Derivations are in the supplementary material (`code/docs/THEORY.md`); their numerical checks are in `results/theory_check_*.csv`.

4 Experiments

4.1 Study design

Graphs. Synthetic graphs (Erdős–Rényi, stochastic block model with 20 blocks and 10% of edges between blocks, Barabási–Albert, Watts–Strogatz with rewiring 0.1; $n = 2{,}000$, mean degree 10) and two public graphs: the SNAP ego-Facebook graph (4,039 nodes; McAuley and Leskovec 2012) and the Facebook100 Stanford3 graph (11,586 nodes, mean degree 98; Traud et al. 2012, Rossi and Ahmed 2015; CC BY-SA). Both are anonymised structure only (ego-Facebook has no stated licence). Synthetic cells use 5 independent graph draws $\times$ 100 assignments, real graphs one graph $\times$ 500 assignments. Sweeps use a default SBM ($n = 3{,}000$, mean degree 20) unless stated, with 8 graph draws $\times$ 250 assignments.

Mechanisms. $Y_i = b_i + \mu_i(\mathbf{z}) + \varepsilon_i$ with a fixed unit baseline b_i (correlation 0.5 with standardised log-degree plus an i.i.d. community component of SD 1, so that position in the graph confounds the outcome) and Gaussian noise (SD 1). M1: $\mu_i = \delta z_i + \gamma f_i$ (linear in the treated-neighbour fraction); M3: $\gamma g(f_i)$ with concave saturating $g(f) = (1 - e^{-3f})/(1 - e^{-3})$; M4: $\delta z_i + 0.5 f_i + 0.5\,(W^2\mathbf{z})_i$ (one- plus two-hop interference); M5: M1 with $\gamma_i = \gamma(1 + 0.5\, s_i)$, s_i the standardised log-degree. Defaults: $\delta = 1$ and total spillover 1 (interference-to-direct ratio $\rho = 1$, so $\tau = 2$ in all four). The main-grid parameters were fixed before any estimator results were seen; sweeps vary one factor at a time (Table 2 and Appendix).

Designs. Bernoulli(0.5), Bernoulli(0.1), and graph-cluster randomisation (Louvain clusters, each cluster treated with probability 0.5).

Estimators. The difference in means (DM), a Hájek exposure estimator (strict full-neighbourhood exposure, with a $q = 0.75$ variant for Bernoulli designs), and OLS on (z, f) (`ols_f`, the Chin-type estimator) with variants adding covariates, degree interactions, a two-hop feature or polynomial terms; ridge and gradient boosting appear in a smaller sub-experiment.

Metrics. Bias, SD and RMSE of $\hat{\tau} - \tau$ over assignments, and RMSE and bias relative to $|\tau|$ (scale-free: "fraction of the effect"). Also the coverage of nominal 95% intervals (cluster-robust sandwich with Louvain communities as clusters, and a heteroskedasticity-robust one treating units as independent) and the share of replicates in which an estimator is undefined. Uncertainty: 95% percentile bootstrap intervals over replicates for every plotted point; estimators share assignments and noise within a replicate (paired comparisons); an independent second seed set for the synthetic main grid is in Appendix B.

4.2 RQ1: how accurate is it?

Figure 1 and Table 1 give the headline. Under Bernoulli(0.5) with linear exposure (M1) the Chin-type OLS has median relative RMSE 0.19 across the six graphs against 0.50 for DM, and it is closer to the truth than DM in every graph of the main grid (Table 1). Among the synthetic graphs, ER and BA (0.14, 0.14) are easier than the SBM and Watts–Strogatz (0.25, 0.25); of the real graphs Stanford3 is easy (0.10) and ego-Facebook (0.34) depends on the baseline draw (Fig. 1; Sec. 4.4). The strict Hájek estimator is no competitor under Bernoulli designs: it is undefined in a median of 0.99 of replicates at $p = 0.1$, and at $p = 0.5$ worse than OLS(z, f) in every graph (0.28–1.23 against 0.10–0.34). Adding a feature the mechanism does not need (the two-hop regressor under M1) is mainly a variance cost and is worse than DM on the SBM (1.30), ego-Facebook (0.87) and Stanford3 (0.72). The advantage is not uniform: under the saturating mechanism M3 at $p = 0.1$ the Chin-type OLS has median relative RMSE 0.63 and median relative bias +0.53, and it loses to DM on five of six graphs; only Stanford3, with its degree of 98, is better (Table 1); two-hop interference (M4) gives a median relative bias of -0.22 at $p = 0.5$, which a two-hop feature removes (+0.01) at a variance price.

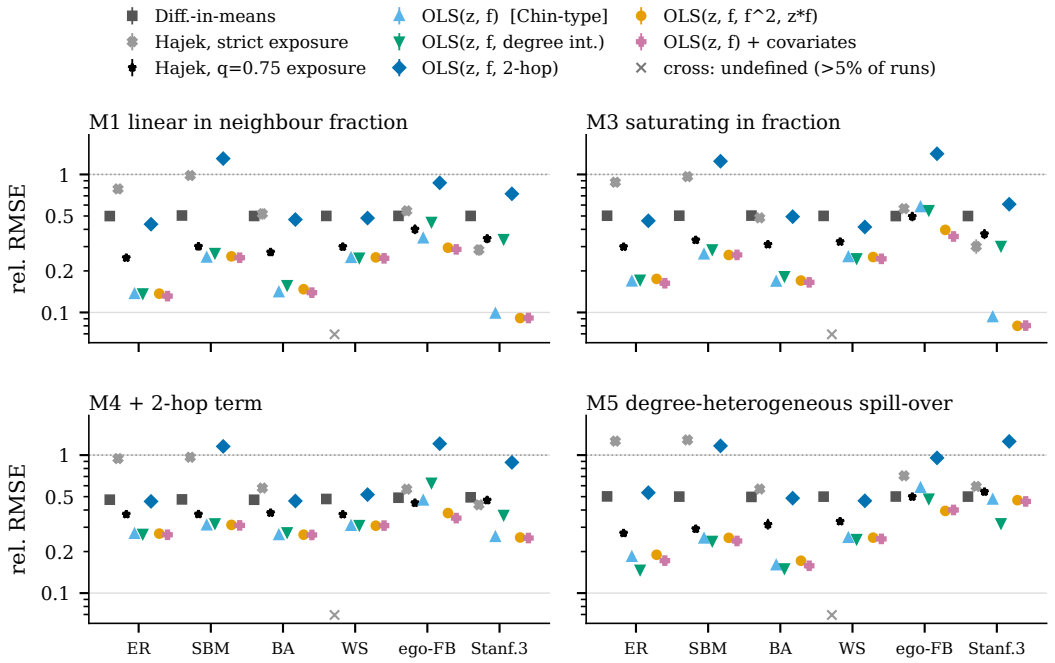


Figure 1: Relative RMSE (RMSE/$|\tau|$, log scale, 95% bootstrap CI) of each estimator on the 24 mechanism×graph cells of the main grid under Bernoulli(0.5), $\rho = 1$. The difference in means has relative RMSE ≈ 0.5 everywhere because it recovers only the direct effect. Cells in which the strict Hájek estimator is undefined in more than 5% of replicates are drawn as crosses. Other designs: Figures 6–7.

Table 1: Main grid, median over six graphs [min, max] of relative RMSE of DM and of OLS(z, f), median relative bias of OLS(z, f), and the number of graphs on which OLS(z, f) is better than / worse than / not significantly different from DM in paired MSE (95% bootstrap CI of the MSE ratio, `results/contrasts.csv`).

mech.	design	rel. RMSE, median [min, max] DM	rel. RMSE, median [min, max] OLS(z, f)	rel. bias OLS(z, f)	OLS better / worse / n.s.
M1	Bernoulli(0.5)	0.50 [0.50, 0.50]	0.19 [0.10, 0.34]	+0.00	6 / 0 / 0
M1	Bernoulli(0.1)	0.50 [0.50, 0.51]	0.31 [0.16, 0.57]	+0.00	5 / 1 / 0
M1	graph-cluster (p=0.5)	0.27 [0.15, 0.37]	0.18 [0.13, 0.25]	+0.00	4 / 1 / 1
M3	Bernoulli(0.5)	0.50 [0.50, 0.50]	0.21 [0.09, 0.58]	-0.09	5 / 1 / 0
M3	Bernoulli(0.1)	0.50 [0.50, 0.51]	0.63 [0.30, 0.80]	+0.53	1 / 5 / 0
M3	graph-cluster (p=0.5)	0.31 [0.18, 0.40]	0.17 [0.16, 0.70]	-0.05	4 / 1 / 1
M4	Bernoulli(0.5)	0.48 [0.47, 0.49]	0.29 [0.26, 0.47]	-0.22	5 / 0 / 1
M4	Bernoulli(0.1)	0.48 [0.47, 0.50]	0.38 [0.28, 0.63]	-0.21	5 / 1 / 0
M4	graph-cluster (p=0.5)	0.33 [0.17, 0.40]	0.22 [0.16, 0.59]	-0.06	3 / 2 / 1
M5	Bernoulli(0.5)	0.50 [0.50, 0.50]	0.25 [0.16, 0.58]	-0.09	5 / 1 / 0
M5	Bernoulli(0.1)	0.50 [0.50, 0.51]	0.42 [0.23, 0.71]	-0.09	4 / 1 / 1
M5	graph-cluster (p=0.5)	0.31 [0.17, 0.38]	0.18 [0.16, 0.44]	-0.09	3 / 1 / 2

4.3 RQ2: what determines the error?

Variance dominates when the model is right, and is set by the design. Under correct specification the estimator is unbiased (|relative bias| ≤ 0.009 for all n) and its SD falls as $n^{-0.51}$ (fitted log–log slope; Fig. 2). Because DM has a constant bias of -0.50 of the effect, OLS crosses it near $n = 2{,}000$: at $n = 500$ the regression is worse than DM (0.86 vs. 0.51), at $n = 8{,}000$ it is better (0.21 vs. 0.50). The absolute SD of OLS is almost independent of the interference strength (0.62–0.76), so its error *relative* to the effect $(1 + \rho)$ falls with ρ (Fig. 2); DM is much better with no interference ($\rho = 0$:

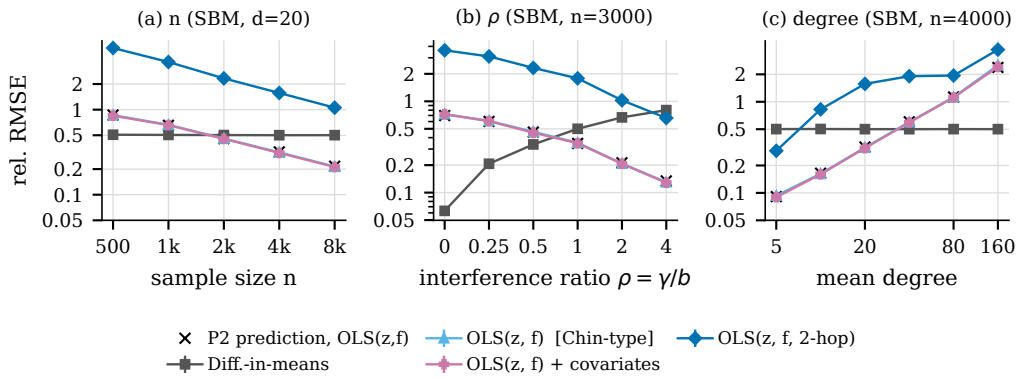


Figure 2: Relative RMSE of DM and OLS variants against (a) n (SBM, mean degree 20), (b) the interference ratio ρ ($n = 3{,}000$) and (c) mean degree ($n = 4{,}000$); 8 graph draws $\times$ 250 assignments per point. Crosses: closed-form P2 standard deviation of OLS(z, f). Covariates change nothing visible; the two-hop regressor is worse except at $\rho = 4$. Bias and SD panels, with the count-feature variants, are in Figs. 16–17.

0.06 vs. 0.72) and still better at $\rho = 0.5$ (0.34 vs. 0.46), and worse from $\rho = 1$ (0.50 vs. 0.35); at $\rho = 4$ OLS has 0.13 against 0.80. Which graph quantities set the variance is predicted by P2: the SD grows with the (harmonic mean) degree, because the regressor f then varies little across units and the coefficient of f is weakly identified, and with the part of the baseline that is smooth over the graph. The predicted SD tracks the simulated one over 496 graph draws (median ratio 1.008, IQR 0.977–1.039; Fig. 23). Concretely, on the SBM the relative RMSE grows from 0.09 at mean degree 5 to 2.51 at 160, where DM would be better; on ER it grows more slowly (0.08 to 0.43, Fig. 14). A community-level component of the baseline raises the error most: with $\sigma_c = 0$ the relative RMSE is 0.12, with $\sigma_c = 3$ it is 0.98 (Fig. 15), which is above DM's 0.5.

Bias comes from extrapolation and specification. The regression is fit where the exposure f is concentrated near the design's p and then evaluated at $f = 0$ and $f = 1$. P1 turns this into a computable asymptotic bias, $\beta_f^* - [g(1) - g(0)]$, which matches the noise-free population fits in 626 graph draws (correlation 0.99998, maximum absolute difference 0.081). Figure 3 shows what it implies: for each smooth response the bias vanishes at one value of p and changes sign around it (for $f^{0.5}$ at degree 10 the relative bias is +0.44 at $p = 0.1$ and -0.23 at $p = 0.9$; for f^2 it is 0.00 at $p = 0.5$ and -0.34, +0.34 at $p = 0.1, 0.9$); for a threshold response at $f = 0.5$ it is +0.34 at $p = 0.5$ and -0.27, -0.30 at $p = 0.1, 0.9$. It is a population quantity, so more data do not reduce it, and higher degree does not either, since it concentrates f around p: the threshold bias at $p = 0.5$ grows from +0.34 (degree 10) to +0.71 (degree 160). Whether a feature set can express the mechanism matters in the same way (Fig. 18): adding a two-hop term to the signal gives a relative bias of -0.23 for the Chin-type OLS on the SBM. The raw neighbour *count* (scaled by mean degree) is confounded with degree and hence with the degree-correlated baseline: on ER with mean degree 60 its relative RMSE is 1.65 when the truth is the fraction and 1.97 when the truth is the count itself, and its relative bias is 2.17 even with no interference ($\rho = 0$, Fig. 17). Controlling for log-degree and clustering removes this on ER (0.29 and 0.28), but on BA with fraction exposure the same adjusted regression has a relative bias of -0.41 (Fig. 18). Heterogeneous spillovers that correlate with degree bias the estimator by P3 (M5 in Table 1).

Flexible learners do not remove it. H2 predicted that tree ensembles, which do not extrapolate, would fail. The evidence is mixed (Fig. 4; fewer replicates: 4 graph draws $\times$ 80 assignments). Under linear exposure, ridge regression on rich features (relative RMSE 0.94 at degree 10, 2.24 at degree 40) and a cubic OLS at degree 40 (3.51) are far worse than OLS(z, f) (0.20, 0.68); boosting (0.56, 0.76) is biased towards zero at degree 40 (relative bias -0.31). Under the threshold response OLS(z, f) is biased (+0.35 and +0.64 at degrees 10 and 40), boosting is nearly unbiased (+0.02, -0.13) and has lower RMSE than OLS(z, f) at degree 40 (0.73 vs. 0.95) but not at degree 10 (0.56 vs. 0.40); the cubic OLS is best at degree 10 (0.29) and worst at degree 40 (3.73). No learner dominates; the bias–variance trade-off moves with the degree.

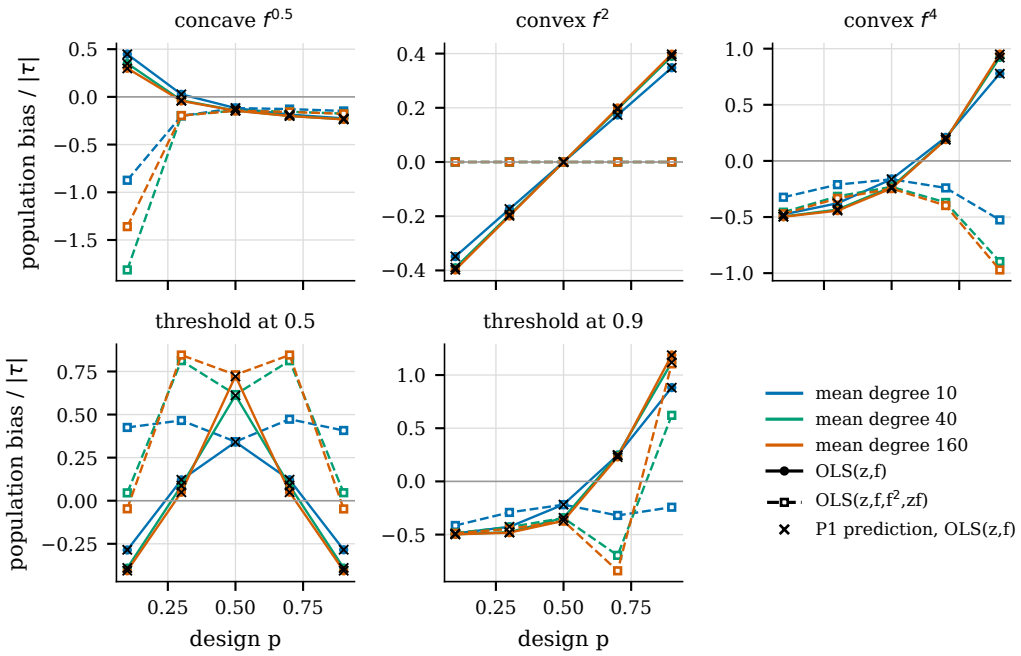


Figure 3: Population (noise- and baseline-free) bias of OLS(z, f) and of the quadratic OLS(z, f, f^2, zf) relative to $|\tau|$, against the design p, for convex (f^2, f^4), concave ($f^{0.5}$) and threshold responses and mean degrees 10, 40, 160 (SBM, $n = 3{,}000$). The estimand is exact; no sampling error enters. OLS(z, f) is unbiased only at special p; the quadratic model removes the bias only when the truth is quadratic.

A computable diagnostic works only for one family (negative result). H7 proposed that a data-computable exposure-support statistic, the distance (in SDs of the realised f) from the design's mean exposure to the farther of $f = 0, 1$, predicts the extrapolation error. Under Bernoulli designs it does for the saturating mechanism ($R^2 = 0.87$ in a log–log regression across cells) but not for power (0.15), threshold (0.05), degree-heterogeneous (0.00) or two-hop (0.00) responses; pooled over all misspecified cells it explains $R^2 = 0.42$ (`results/h7.csv`; Fig. 11 in the appendix shows the main grid). Within the ten cluster-randomised cells per mechanism the statistic is high for saturating (0.89), two-hop (0.99) and heterogeneous (0.68) responses, but these are ten cells per mechanism that differ across graph families and not across p, so we do not weigh them against the Bernoulli result. We do not recommend the statistic as a general warning signal: it tells how far the extrapolation is, not whether the response is curved there.

4.4 RQ3: where does it hold, where does it fail?

Design. Graph-cluster randomisation changes the conclusion: the median relative RMSE of DM falls from 0.50 to 0.27–0.33 across mechanisms (range 0.15–0.40) because exposure is more often all-or-nothing, so the regression's advantage over DM under Bernoulli designs (0.19 vs. 0.50) largely disappears (Fig. 12); OLS is still better than DM in 14 of the 24 cluster-design graph–mechanism pairs (worse in 5, not distinguishable in 5; Table 1), with OLS medians 0.17–0.22 against DM's. It also reduces extrapolation bias (M4: -0.22 → -0.06 in Table 1); for M3 the median OLS error under cluster randomisation (0.17) is close to that under the linear mechanism (0.18). A small p is costly: variance rises as $1/\sqrt{p(1-p)}$ and nonlinear responses are extrapolated further (Table 1, Bernoulli(0.1) rows).

Network structure. OLS(z, f) beats DM on 9 of ten graph families (relative RMSE 0.11–0.34 against DM's 0.5) but not on ego-Facebook, where in this draw it is worse (0.55; Fig. 20). The main grid and E5 use one draw of the synthetic baseline per real graph, and the same ego-Facebook specification gives a relative RMSE of 0.34 in the main grid, 0.46 in E11 and 0.55 here, so we reran the 24 real-graph main cells with 8 independent baseline draws (60 assignments each). Under M1

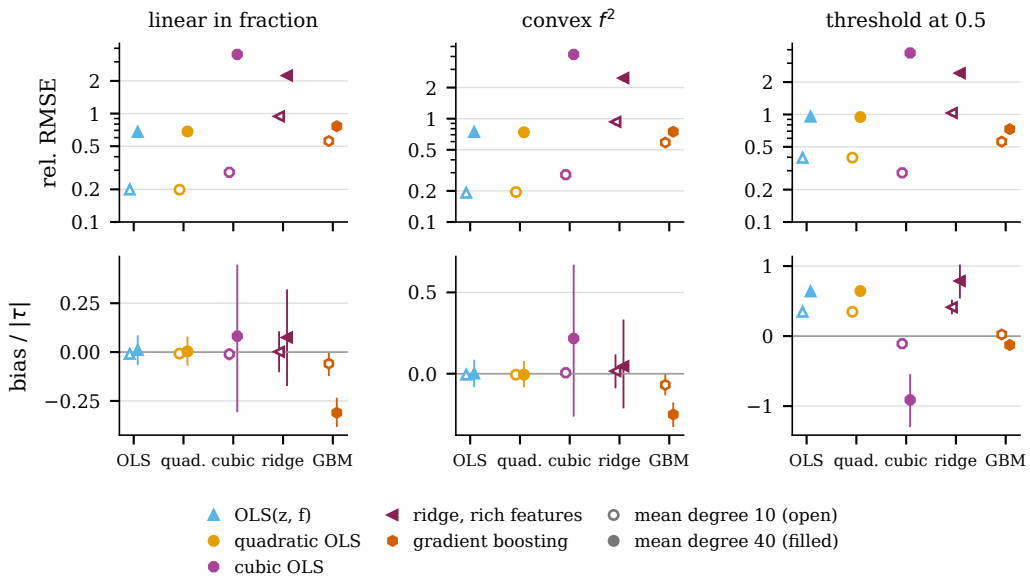


Figure 4: Model class (OLS, quadratic and cubic OLS, ridge on rich features, gradient boosting) under linear, convex and threshold exposure (SBM, $n = 3{,}000$, $p = 0.5$; open: mean degree 10, filled: 40). Relative RMSE and bias with bootstrap CIs; 4 graph draws $\times$ 80 assignments.

and Bernoulli(0.5) the relative RMSE of OLS(z, f) on ego-Facebook has mean 0.43 over the draws (range 0.27–0.65, SD 0.11) against 0.50 for DM, and on Stanford3 0.12 (0.09–0.16). Over the four mechanisms at $p = 0.5$ OLS beats DM in 32 of 32 draw–mechanism pairs on Stanford3 (31 of 32 at $p = 0.1$, 15 under cluster randomisation) but in only 21 of 32 on ego-Facebook (1 of 32 at $p = 0.1$, 10 under cluster randomisation), with the saturating and degree-heterogeneous mechanisms the worst (4 and 2 of 8). So Stanford3 favours regression robustly, whereas on ego-Facebook the answer depends on the baseline draw and DM is competitive; the count of families above is a single-draw count for ego-Facebook. Degree is the main structural driver (above); clustering and block structure matter through the baseline projection in P2 (at equal mean degree 20 and $n = 4{,}000$ the relative RMSE is 0.13 on ER, 0.23 on the SBM with 40% of edges between blocks and 0.34 with 2%; Fig. 20).

Graph errors. Dropping and re-adding a share of edges when building the features biases the estimate towards zero, monotonically in the error rate, by $-0.26|\tau|$ at 50% on the SBM and $-0.27|\tau|$ on ego-Facebook (Fig. 13). At $\rho = 1$ the relative RMSE rises from 0.21 to 0.31 on the SBM (DM: 0.50) and stays below DM, whereas on ego-Facebook it is already 0.46 without errors and 0.43 at 50% (about DM's level). With weak interference ($\rho = 0.25$) the regression with an error-free graph already has relative RMSE 0.38 against 0.21 for DM, so DM wins at every error rate on both graphs.

Inference. In the correctly specified linear mechanism the cluster-robust 95% interval covers between 0.85 and 0.94 across graph families and sizes, below the nominal 95% although the bias is negligible: on the SBM it improves with n (0.88 at $n = 1{,}000$ to 0.94 at $4{,}000$), on BA it worsens (0.90 to 0.85). The interval that treats units as independent covers only 0.56 at $n = 1{,}000$ and 0.50 at $4{,}000$ when baselines are community-structured and mixing is weak (0.73 and 0.65 for strong mixing), so its under-coverage grows with n; it covers 0.87–0.92 on ER and BA (Fig. 5). On the main grid coverage collapses once $|\text{bias}|/\text{SD} > 1$ (Fig. 10), as expected for an interval centred on a biased estimate. How coverage changes with n when the model is misspecified (the second part of H6) was not tested.

5 Discussion and Limitations

What the evidence says. Against the seven hypotheses we pre-specified (Table 3, Appendix), H1, H3 and H5 are supported, H2, H4 and H6 partly, and H7 not as a general law. Regression adjustment

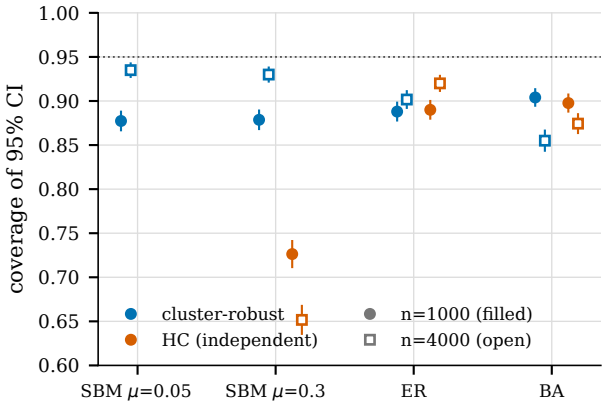


Figure 5: Coverage of the nominal 95% interval for OLS(z, f), cluster-robust versus independent-units standard errors, $n = 1{,}000$ (filled) and $4{,}000$ (open), correct linear mechanism, 10 graph draws $\times$ 300 assignments.

for the GATE is a bias–variance bargain whose terms are set by the design and the graph, not by the sample size alone. When the exposure model is right it is the most accurate of the estimators we compared once n and the interference are large enough (in the default SBM setting, $n \gtrsim 2{,}000$ and interference at least as strong as the direct effect; the break-even moves with degree and confounding); under Bernoulli(0.5) its median relative RMSE over the six main-grid graphs is then 0.19 (DM: 0.50). It is not a free lunch: its variance grows with degree and with community-level baseline confounding, and a difference in means is better with weak interference, small n, or after cluster randomisation. When the exposure model is wrong, the error is a bias that more data do not remove; we can compute its first-order size in the linear family (P1) and observe it for nonlinear, two-hop, degree-heterogeneous and miswired responses.

Practical reading. (i) Fit the simplest exposure model, but check the design's exposure support: bias is largest where f is far from the design's mean exposure. (ii) Prefer balanced designs; at $p = 0.1$ nonlinear bias and variance both grow. (iii) Do not rely on high-capacity learners to escape misspecification. (iv) Treat nominal 95% intervals, cluster-robust ones included, as optimistic. These are readings of simulations.

Limitations. *Synthetic outcomes.* The graphs are real or generated, but every outcome is simulated from mechanisms we chose; we did not test the estimators on any experiment with real outcomes, and our mechanisms (linear, saturating, power, threshold, two-hop, degree-heterogeneous) are a small family. The relative scale (e.g. DM at 0.5) is a function of the ratio ρ we chose. *Replication.* Synthetic cells use 5–8 graph draws; each real graph is one graph, and its baseline draw, which the bootstrap does not cover, matters (Sec. 4.4). Sweeps vary one factor at a time. The model-class comparison (Fig. 4) uses 4 graph draws $\times$ 80 assignments, and a ridge/boosting evaluation on the full main grid was not completed (an attempt was stopped; Appendix A). *Analytics.* P1–P3 are first-order large-n arguments for the linear-exposure family, checked numerically, not proved, and do not cover boosted or cubic fits. *Inference.* Louvain communities are both the cluster design and, in some cells, the variance clusters, which favours the cluster-robust interval; we studied no corrections. *Not done.* Multi-p and staggered designs (H4), the dependence of coverage on n under misspecification (H6), unknown-interference-radius selection, and optimal-design recommendations. The H7 diagnostic did not generalise, so we offer no data-driven warning signal.

References

Peter M. Aronow and Cyrus Samii. Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics*, 11(4):1912–1947, 2017.

Qianyi Chen, Bo Li, Lu Deng, and Yong Wang. Optimized covariance design for AB test on social network under interference. In *Advances in Neural Information Processing Systems 36*, 2023.

Alex Chin. Regression adjustments for estimating the global treatment effect in experiments with interference. *arXiv preprint arXiv:1808.08683*, 2019.

Mayleen Cortez, Matthew Eichhorn, and Christina Lee Yu. Staggered rollout designs enable causal inference under interference without network knowledge. In *Advances in Neural Information Processing Systems 35*, 2022.

Dean Eckles, Brian Karrer, and Johan Ugander. Design and analysis of experiments in networks: Reducing bias from interference. *Journal of Causal Inference*, 5(1), 2017.

Kevin Han and Johan Ugander. Model-based regression adjustment with model-free covariates for network interference. *arXiv preprint arXiv:2302.04997*, 2023.

Brian Karrer, Liang Shi, Monica Bhole, Matt Goldman, Tyrone Palmer, Charlie Gelman, Mikael Konutgan, and Feng Sun. Network experimentation at scale. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2021.

Michael P. Leung. Rate-optimal cluster-randomized designs for spatial interference. *The Annals of Statistics*, 50(5), 2022.

Julian McAuley and Jure Leskovec. Learning to discover social circles in ego networks. In *Advances in Neural Information Processing Systems 25*, 2012.

Ryan A. Rossi and Nesreen K. Ahmed. The network data repository with interactive graph analytics and visualization. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, 2015.

Fredrik Sävje, Peter M. Aronow, and Michael G. Hudgens. Average treatment effects in the presence of unknown interference. *The Annals of Statistics*, 49(2), 2021.

Amanda L. Traud, Peter J. Mucha, and Mason A. Porter. Social structure of Facebook networks. *Physica A: Statistical Mechanics and its Applications*, 391(16):4165–4180, 2012.

Johan Ugander, Brian Karrer, Lars Backstrom, and Jon Kleinberg. Graph cluster randomization: network exposure to multiple universes. In *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 329–337, 2013.

Christina Lee Yu, Edoardo M. Airoldi, Christian Borgs, and Jennifer T. Chayes. Estimating the total treatment effect in randomized experiments with unknown network structure. *Proceedings of the National Academy of Sciences*, 119(44), 2022.

A Setup and reproduction

Table 2 summarises the simulation grid. All code, the data-acquisition steps with checksums, seeds, commands and the figure/table-to-script map are in the supplementary material (`code/README.md`, `figures/README.md`); every number quoted in the text is generated into `paper/numbers.tex` by `paper/scripts/make_numbers.py` from `results/`. Base seed 20261102. The 24 real-graph cells of the main grid were rerun with 8 independent baseline draws $\times$ 60 assignments (experiment `MAIN_realdraws`, Sec. 4.4). No material excluded by the task was consulted.

Estimator details. All OLS variants use ordinary least squares on the listed features plus an intercept ("covariates" are standardised log-degree and local clustering coefficient). The ridge regression uses the nine features $z, f, (W^2\mathbf{z}), f^2, zf, \ell, f\ell, z\ell, c$ and the scaled neighbour count (ℓ standardised log-degree, c clustering), standardised, with the penalty chosen by leave-one-out cross-validation over 13 values in $[10^{-3}, 10^{3}]$. Gradient boosting is scikit-learn's `HistGradientBoostingRegressor` (200 iterations, learning rate 0.1, at most 15 leaves, fixed seed) on $(z, f, W^2\mathbf{z}, \ell, c)$. Neither learner is tuned per cell. DM is the treated–control difference of means; the strict Hájek estimator contrasts treated units whose neighbours are all treated with control units whose neighbours are all control, weighted by inverse design probabilities (undefined if either group is empty); the $q = 0.75$ version relaxes "all" to at least 75% of neighbours and is implemented for Bernoulli designs only.

Compute. All runs are CPU-only, in containers limited to 2 CPUs and 4 GB on a shared host (Xeon Gold 6230, load about 30), so the wall times below are upper bounds for an idle machine; one cell runs per process on 2 processes. Measured wall time: the main grid with the validity cells (76 cells) 21 min; the second-seed rerun (48 cells) 18 min; sweeps E1–E3, E5, E6, E8–E10 together 95 min; E4 (90 cells) 53 min; E11 about 2.5 min; analytic checks 38 min; export 32 s. The model-class experiment E7 (ridge and boosting) took 21 h because the learners oversubscribed the host's threads; with threads limited, a pilot needs about 0.4 s per replicate. A first attempt at the same learners on the full main grid was stopped after 38 h and is not reported. Peak memory is about 1.5 GB per process for Stanford3 and 3.5 GB for the export; per-group memory was not otherwise measured. The experiment table and the commands are in `code/README.md`; the code regenerates the delivered cell specifications, and a check script re-runs a cell and compares it with the stored replicates (identical for the cells we checked, including the main grid).

Table 2: Setup of the main grid.

Graphs (size n, mean degree $\bar{d}$, replicates per cell)	
ER	$n = 2{,}000$, $\bar{d} = 10.0$, 5 graph draw(s) $\times$ 100 assignments
SBM	$n = 2{,}000$, $\bar{d} = 10.0$, 5 graph draw(s) $\times$ 100 assignments
BA	$n = 2{,}000$, $\bar{d} = 10.0$, 5 graph draw(s) $\times$ 100 assignments
WS	$n = 2{,}000$, $\bar{d} = 10.0$, 5 graph draw(s) $\times$ 100 assignments
ego-FB	$n = 4{,}039$, $\bar{d} = 43.7$, 1 graph draw(s) $\times$ 500 assignments
Stanford3	$n = 11{,}586$, $\bar{d} = 98.1$, 1 graph draw(s) $\times$ 500 assignments
Mechanisms	M1 linear in neighbour fraction f_i; M3 saturating $g(f_i)$; M4 = M1 + 2-hop term; M5 = M1 with degree-dependent spill-over. The GATE τ is computed exactly from $m(\mathbf{1}) - m(\mathbf{0})$ (no rescaling); it equals 2 in all four mechanisms (direct effect 1, total spill-over 1).
Designs	Bernoulli(0.5), Bernoulli(0.1), graph-cluster randomisation (Louvain, $p = 0.5$).
Estimators	difference-in-means; Hajek exposure (strict and q=0.75); OLS on (z, f), $(z, f,$ degree int.$)$, $(z, f,$ 2-hop$)$, (z, f, f^2, zf), (z, f) + covariates.
Metrics	bias, SD, RMSE, RMSE/$\lvert\tau\rvert$ with bootstrap CIs over replicates; coverage of a cluster-robust 95% interval.

B Additional results

Hypotheses and verdicts. Table 3 states each pre-specified hypothesis, the verdict from the evidence above, and its scope.

Table 3: Pre-specified hypotheses and verdicts. "Scope": what was varied.

	Hypothesis (short)	Verdict (evidence)
H1	Correct spec.: unbiased, $n^{-1/2}$, beats DM more as ρ grows	**Supported**, with a caveat: DM wins for small n or ρ (Fig. 2)
H2	Misspecification bias persists with n; flexible learners do not fix it	**Partly supported**: bias is n-independent (P1, Fig. 3); learners trade bias for variance, boosting helps under a threshold (Fig. 4)
H3	Degree and clustering change the error; cluster designs help	**Supported**: degree sets the variance (P2); cluster designs reduce bias and DM's error; the extra variance expected from fewer independent units was not visible in the median error (Table 1)
H4	Balanced p better than small p; multi-p design helps	**Supported for** $p = 0.5$ vs. 0.1; multi-p/staggered designs **not tested**
H5	Graph errors degrade accuracy monotonically	**Supported for bias** (towards zero, monotone in the error rate, Fig. 13); RMSE is not monotone on ego-Facebook, and at weak interference DM wins even without error
H6	Intervals under-cover when bias matters, worse with n	**Partly**: under-coverage at ≈ 0 bias; the independent-units interval worsens with n in community-structured graphs, the cluster-robust one improves on SBM but not on BA; collapse when $\lvert\text{bias}\rvert/\text{SD} > 1$; the n-trend under misspecification **not tested**
H7	A computable support statistic predicts the extrapolation error	**Not supported in general** (R^2 0.42 pooled; high only for the saturating family)

Paired contrasts over all sweeps (Table 4). For every experiment, each cell of the first-named estimator is compared with the second on the same draws and assignments; entries count cells where its MSE is smaller / not distinguishable / larger (95% paired bootstrap intervals). OLS beats DM in most main-grid cells (50 smaller, 15 larger of 72) but loses in 64 of 90 curvature cells (E4); the two-hop regressor is worse than OLS(z, f) in every main-grid cell, and OLS(z, f) is never better than OLS with covariates in the main grid (worse in 32 of 72 cells).

Table 4: Paired MSE contrasts (smaller / n.s. / larger) of the first estimator against the second, per experiment.

	OLS vs DM	OLS vs Hajek	OLS vs quadr.	OLS vs OLS+cov.	2-hop vs OLS	GBM vs OLS	ridge vs OLS
main grid	50/7/15	31/1/2	38/24/10	0/40/32	0/0/72	–	–
main grid, seed 2	38/6/4	17/0/0	27/20/1	0/36/12	0/1/47	–	–
E1 size n	3/0/2	–	1/4/0	0/3/2	0/0/5	–	–
E2 degree	9/0/3	–	7/5/0	0/1/11	0/0/12	–	–
E3 ρ	3/0/3	–	1/5/0	0/3/3	0/0/6	–	–
E4 curvature $\times p$	24/2/64	33/2/1	78/12/0	1/70/19	–	–	–
E5 graph families	9/0/1	–	4/4/2	0/3/7	0/0/10	–	–
E6 specification	15/0/0	–	11/4/0	0/4/11	0/0/15	–	–
E7 model class	–	–	1/5/0	–	–	1/1/4	0/0/6
E8 design	10/8/6	–	15/9/0	1/18/5	–	–	–
E9 noise	5/1/3	–	3/6/0	0/4/5	0/0/9	–	–
E10 coverage	–	–	–	0/4/12	–	–	–
E11 graph error	7/5/12	–	1/18/5	0/11/13	–	–	–

Other designs (Fig. 6, 7). The headline figure for Bernoulli(0.1) and cluster randomisation. At $p = 0.1$ the strict Hájek estimator is undefined in almost all replicates (0.99); at $p = 0.5$ in 0.01.

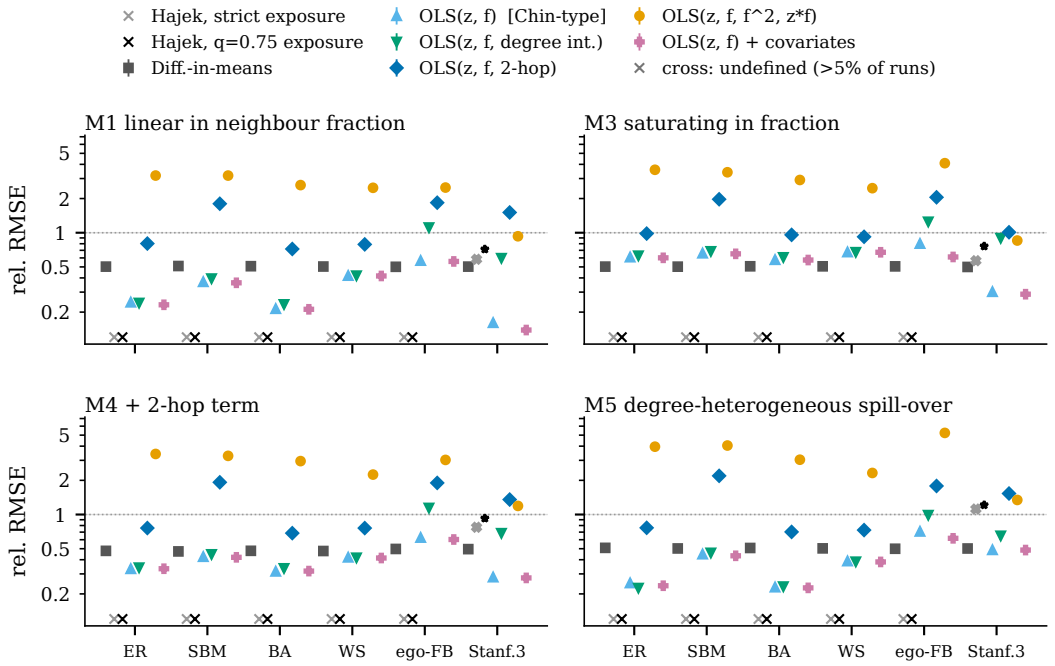


Figure 6: As Fig. 1, Bernoulli(0.1).

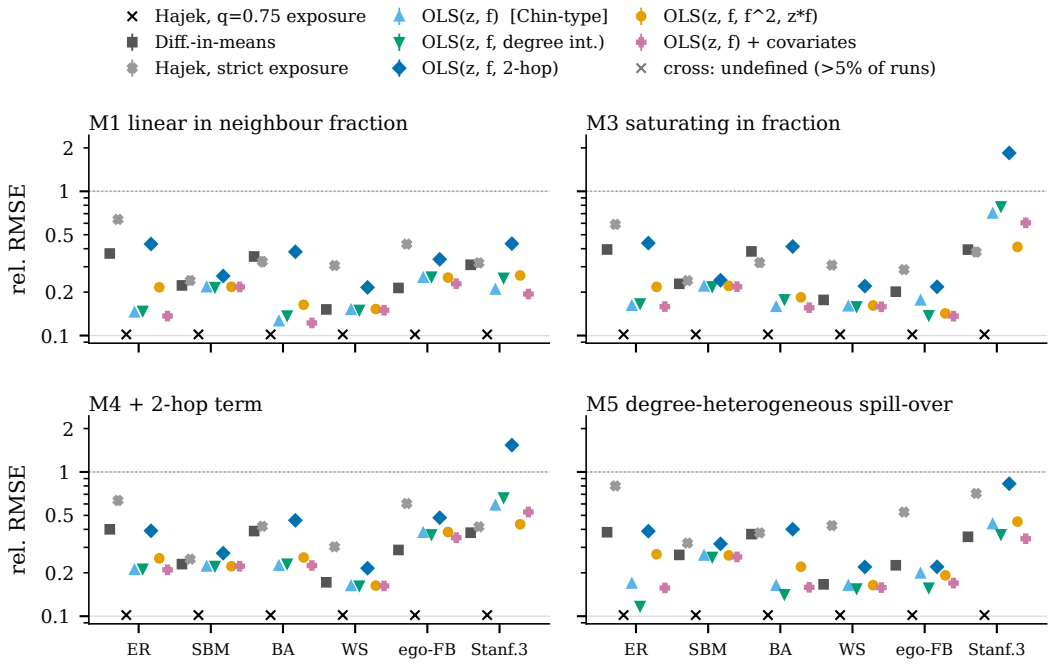


Figure 7: As Fig. 1, graph-cluster randomisation ($p = 0.5$).

Seed replication (Fig. 8). A second, independent seed set for the 48 synthetic main-grid cells reproduces the relative RMSE of every estimator.

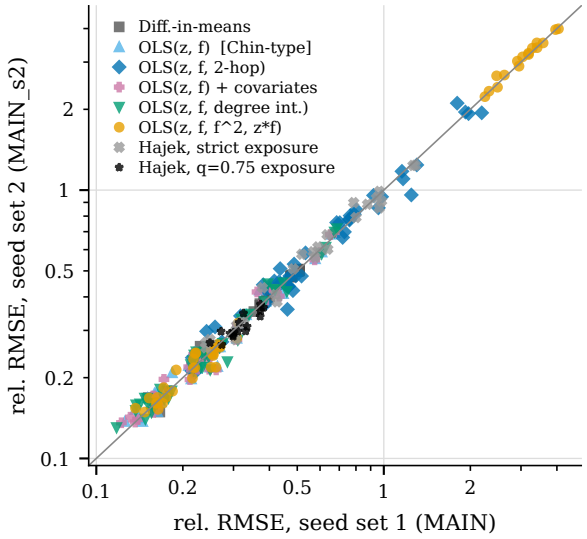


Figure 8: Relative RMSE under the first versus the second seed set, synthetic main-grid cells.

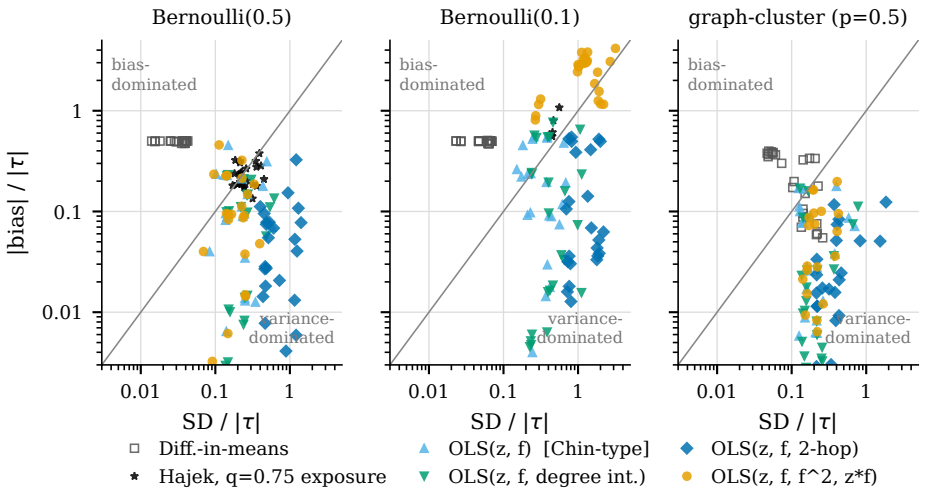


Figure 9: $|\text{Bias}|/|\tau|$ against $\text{SD}/|\tau|$ of each estimator, one point per main-grid cell, by design; the diagonal separates bias- from variance-dominated error.

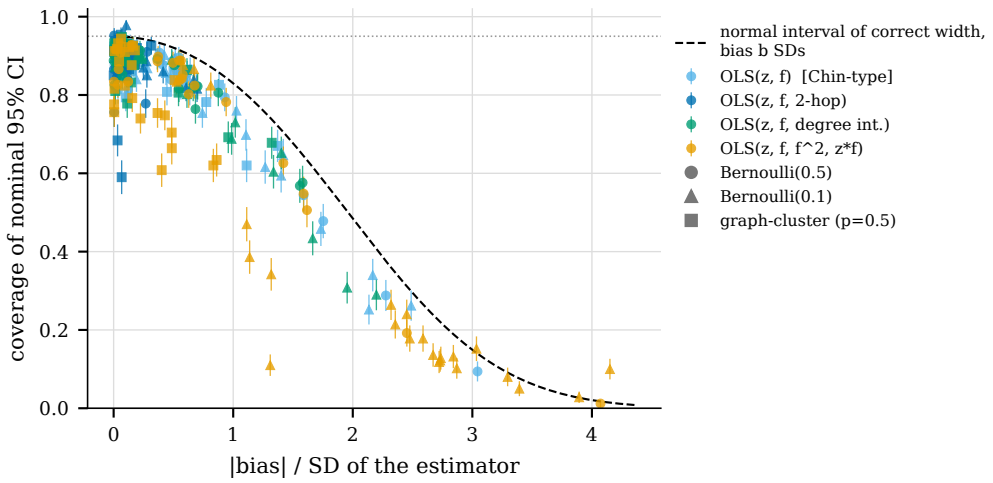


Figure 10: Coverage of the nominal 95% interval against $|\text{bias}|/\text{SD}$, main grid.

384 **Bias versus variance and coverage on the main grid (Figs. 9, 10).**

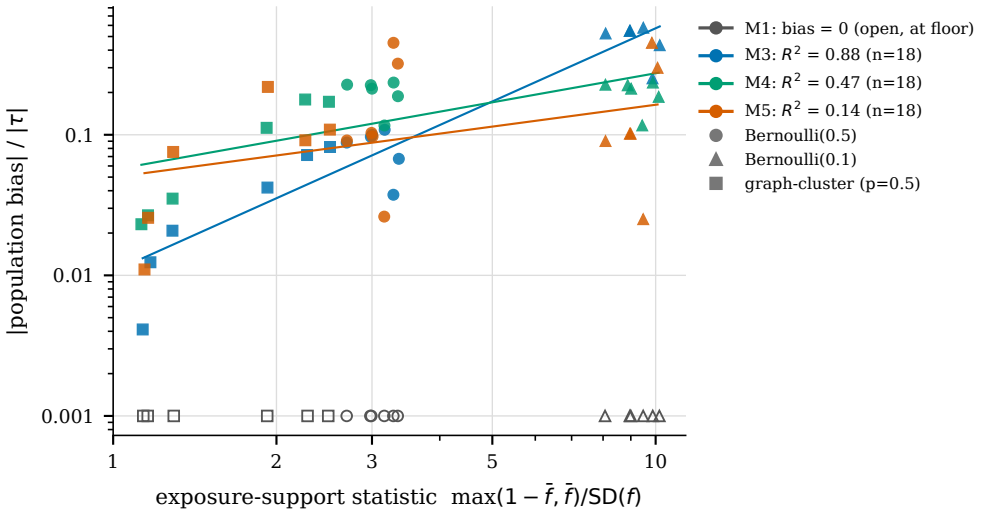


Figure 11: Exposure-support statistic against the absolute population bias of OLS(z, f) on the main grid, one point per cell, per-mechanism log–log regression; cells with no bias (M1) are drawn open at the floor.

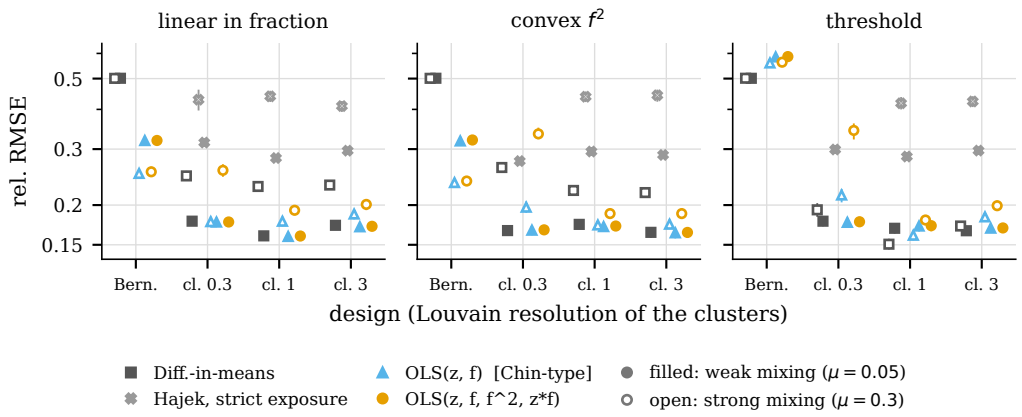


Figure 12: Bernoulli versus graph-cluster randomisation at three Louvain resolutions, weak (filled) and strong (open) mixing between blocks, for linear, convex and threshold exposure (SBM, $n = 4{,}000$, mean degree 20).

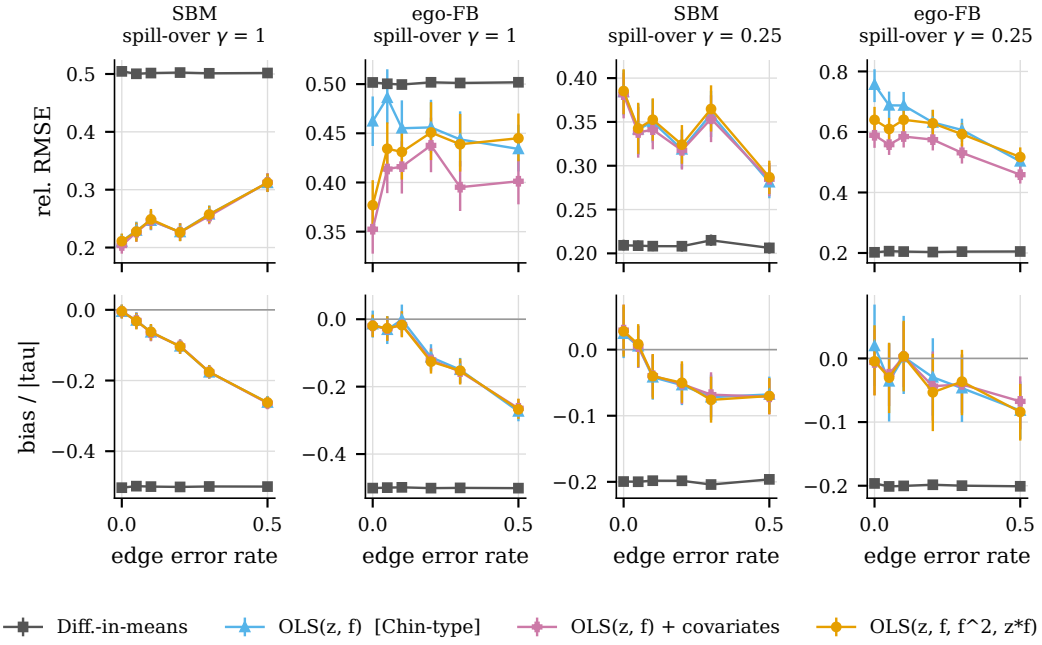


Figure 13: Error against the share of edges dropped and re-added in the graph used to build f (8 perturbation draws on the SBM, 12 on ego-Facebook), strong ($\rho = 1$) and weak ($\rho = 0.25$) interference.

Diagnostic, design and graph errors (Figs. 11, 12, 13).

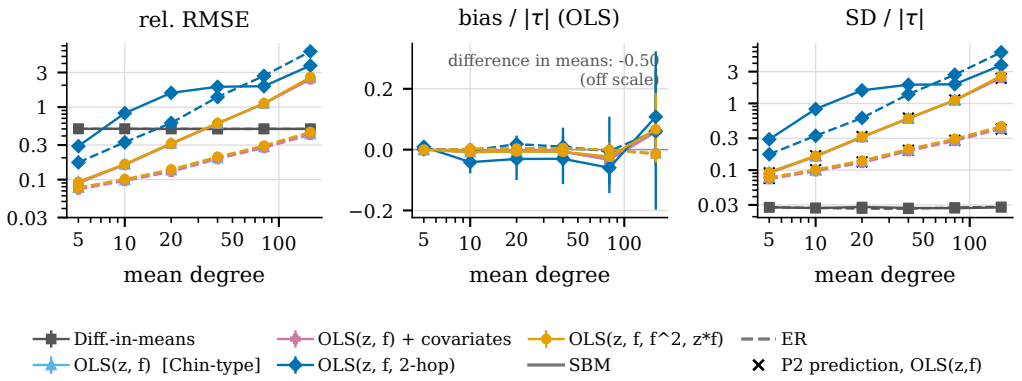


Figure 14: Relative RMSE, relative bias and relative SD against mean degree for the Chin-type OLS(z, f), its degree-interaction and quadratic variants and DM, on SBM (solid) and ER (dashed) graphs.

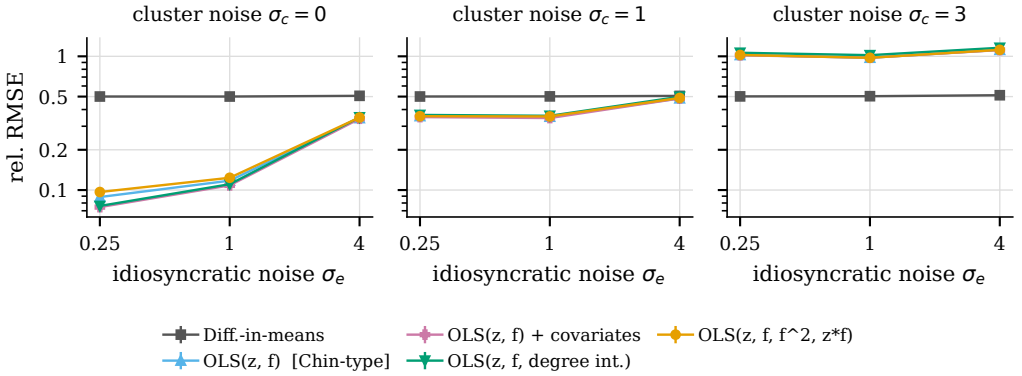


Figure 15: Relative RMSE against the idiosyncratic noise SD σ_e for community-level baseline SD $\sigma_c = 0, 1, 3$ (SBM; DM grey).

386 **Degree and noise (Figs. 14, 15).**

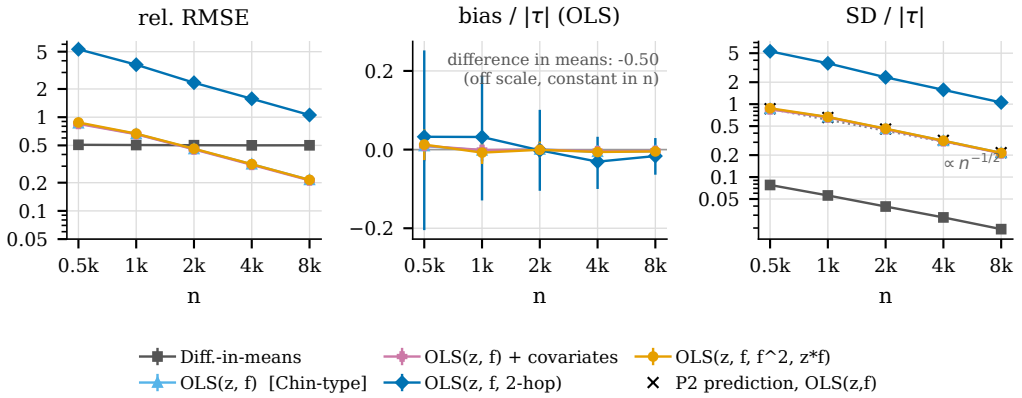


Figure 16: Bias, SD and relative RMSE against n for the correctly specified linear mechanism (SBM, mean degree 20); the SD of OLS follows the $n^{-1/2}$ reference and the P2 prediction.

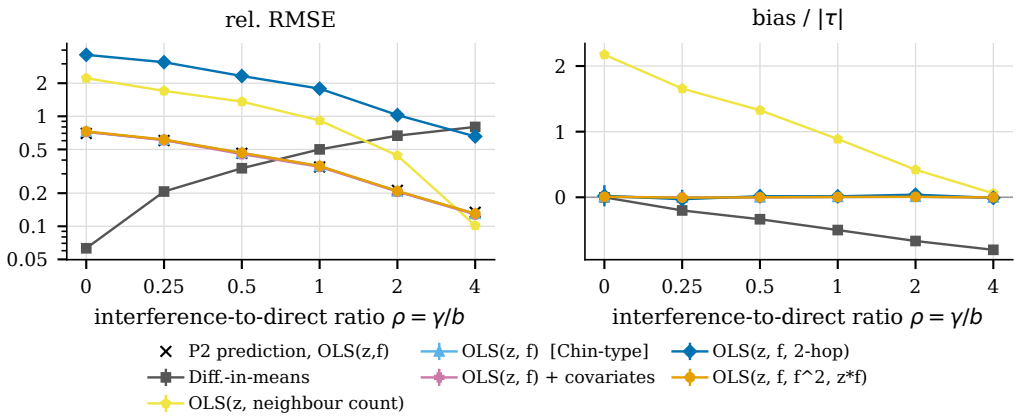


Figure 17: The same against the interference ratio ρ ($n = 3{,}000$). OLS on the raw neighbour count is strongly biased, most at $\rho = 0$.

Bias and SD against n and ρ (Figs. 16, 17).

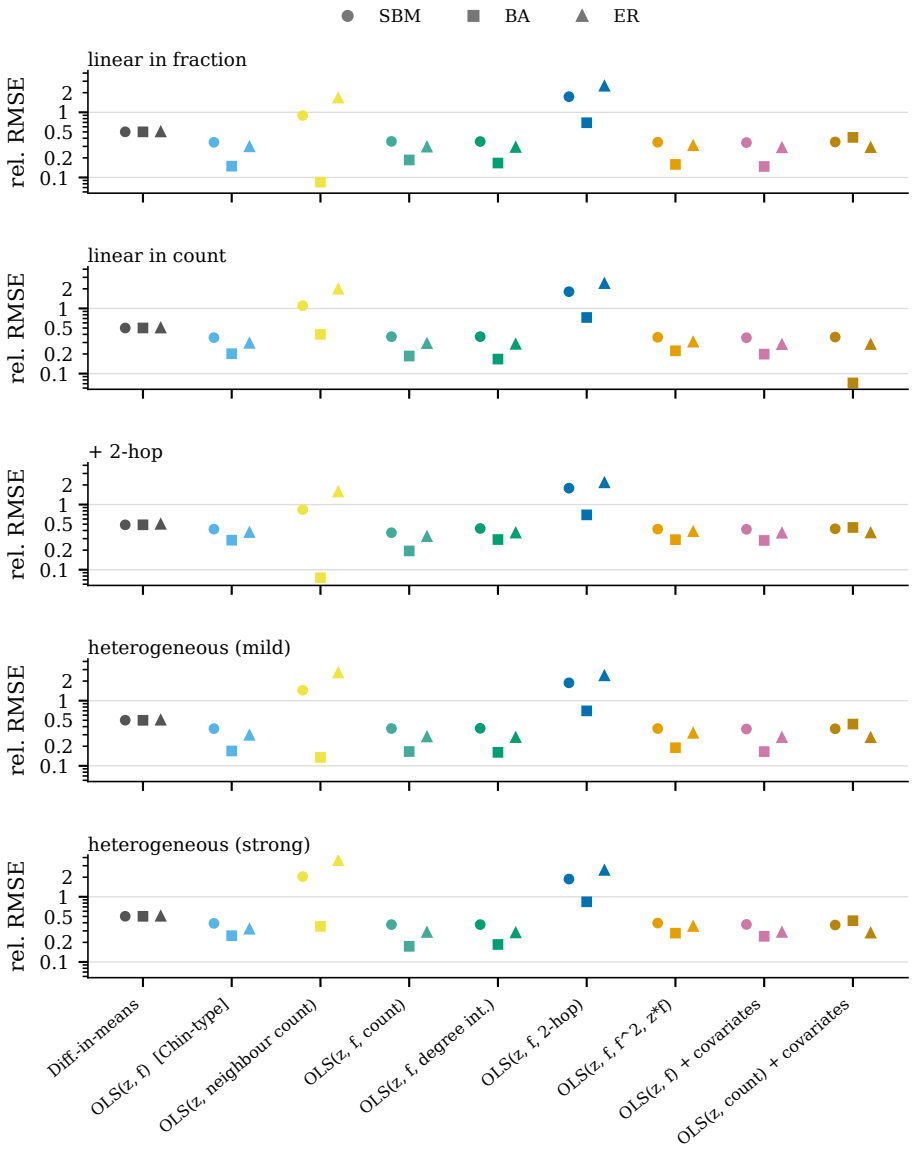


Figure 18: Error of feature sets (rows) for different true exposure forms (linear fraction, linear count, two-hop, degree-heterogeneous) on ER, SBM and BA.

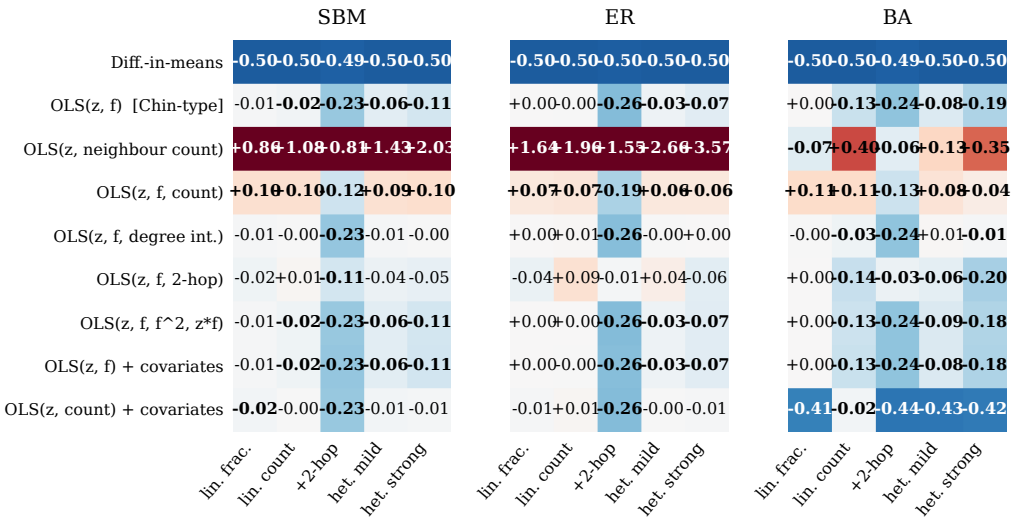


Figure 19: Signed relative bias of the feature sets for each true exposure form (heat map; bold: interval excludes zero). OLS(z, f) is within ± 0.02 on SBM and ER for the linear forms and about -0.25 under an omitted two-hop term; the degree interaction removes the heterogeneity bias but not the two-hop bias.

Feature set versus form of interference (Figs. 18, 19).

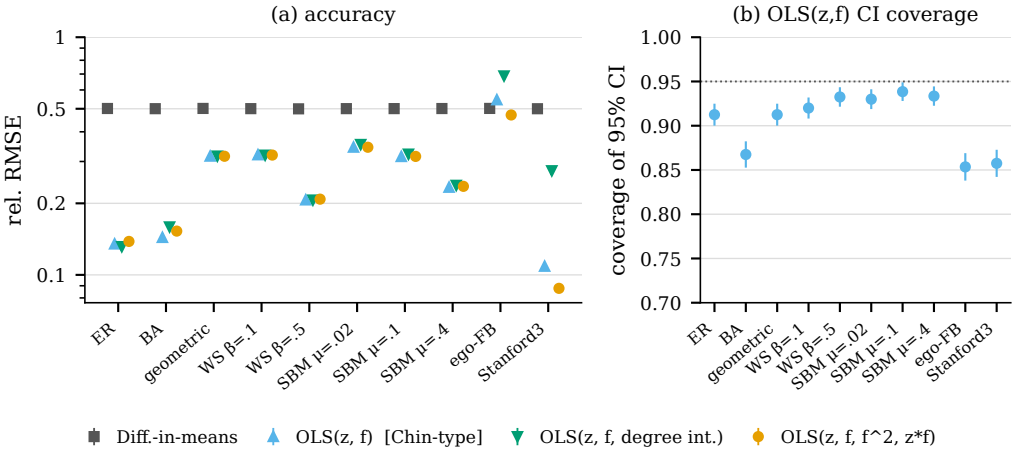


Figure 20: Ten graph families (synthetic at $n = 4{,}000$, mean degree 20, and the two Facebook graphs): (a) relative RMSE of DM and of OLS variants; (b) coverage of the 95% interval for OLS(z, f).

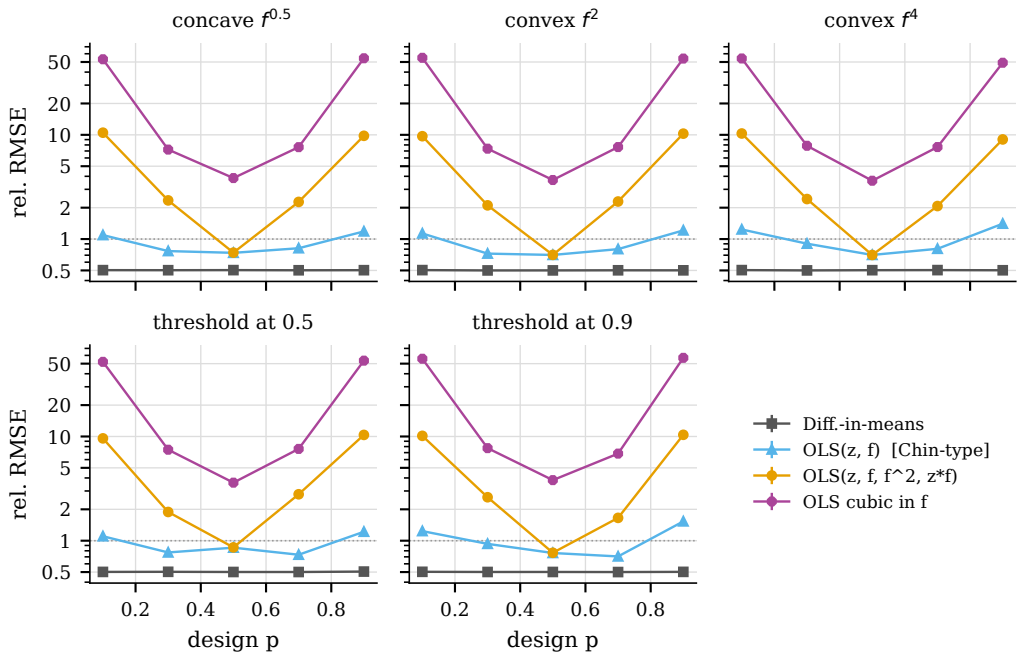


Figure 21: Observed relative RMSE against the design p for the responses of Fig. 3 (degree 40); the quadratic model pays in variance what it gains in bias.

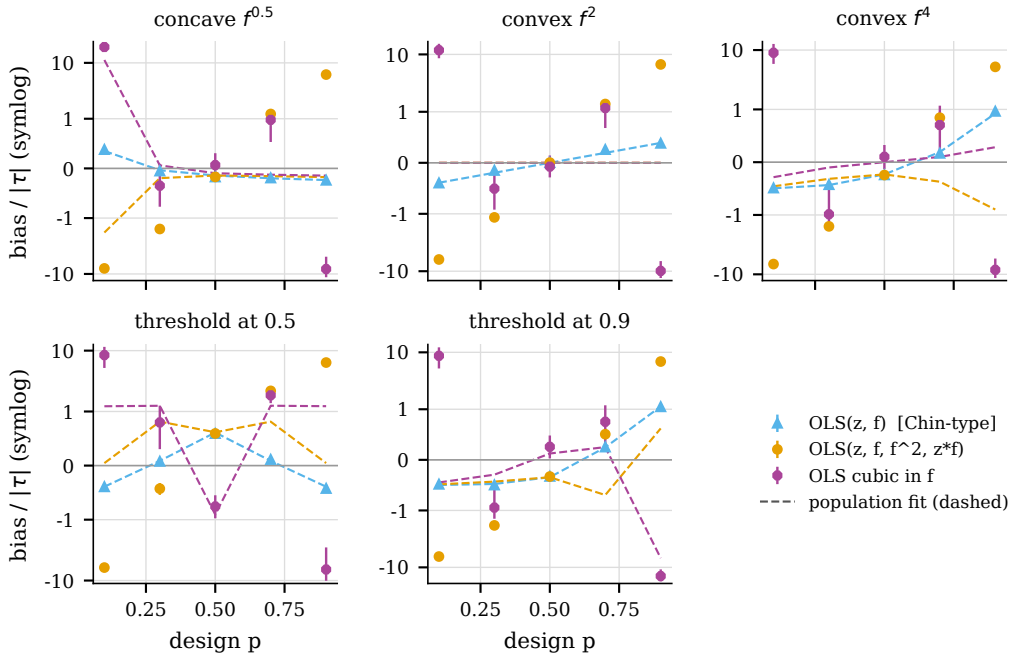


Figure 22: Observed against population-fit bias (degree 40). For OLS(z, f) the two agree; for the quadratic and cubic models the observed bias is far larger at $p = 0.1, 0.9$, i.e. they err by more than their approximation error (we did not test the cause).

Graph families (Fig. 20) and observed error against p.

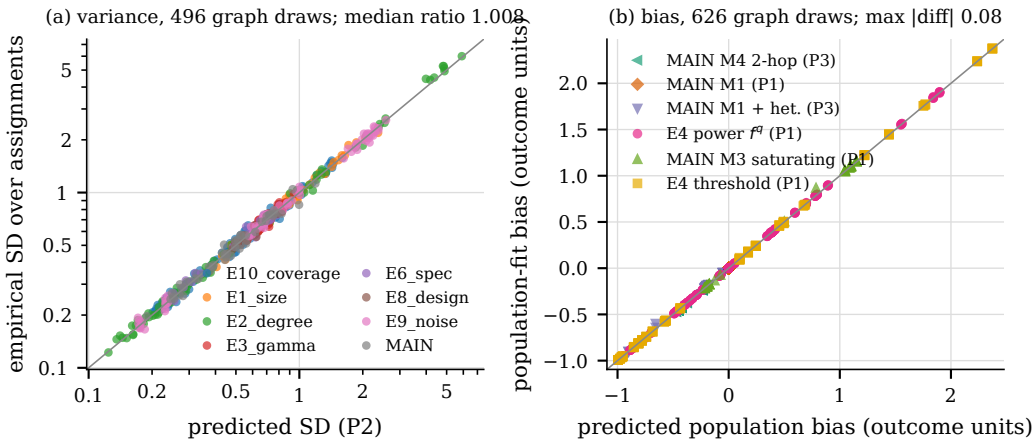


Figure 23: (a) Predicted (P2) against empirical SD of OLS(z, f) over 496 graph draws from several experiments; (b) predicted (P1/P3) against noise-free population bias over 626 draws; both relative to $|\tau|$, diagonal is exact agreement.

Theory checks (Fig. 23).

NeurIPS Paper Checklist

1. **Claims**

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